



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	60.0	☒ 0.0	☒ 0.0	°C
p1	5.0	0.0	0.0	MPa(g)
☐ p2	2.0			MPa(g)
☒ Δp	3.0	☒ 0.0	☒ 0.0	MPa
Cv	20.931	☒	☒	GPM(US)
☐ qm	98,337.0			kg/h
☒ qv	100.0	☒ 0.0	☒ 0.0	m³/h
s/s100	100.0			%
LpAe	91.8			dB(A)
<i>Incipient cavitation</i>				

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	983.37	☒ 0.0	☒ 0.0	kg/m³
ρ2	983.37			kg/m³
☒ η1	0.466	☒ 0.0	☒ 0.0	mPa s
☐ ν1	0.47388			mm²/s
pv1	0.19946	☒	☒	bar(a)
pv2	0.20509			bar(a)
tv1	265.2	99.974	99.974	°C
cF1	1,560.2			m/s





Calculation Sheet

Pipe downstream of valve

Size class downstream of valve	NPS2	2"
Schedule downstream	SCH2	STD

Valve configuration

Valve type	Rotary plug valve
Trim type	Eccentric plug valve
Flow direction	FTC
Valve performance class	Multi stage valve
Protection	Non-hardened
Low-noise design	n.a.

Valve data

Basic characteristic		Equal percentage	
Nominal flow coefficient	Cv100	20.0	GPM(US)
☐ Lift stopper			
Selected valve size	NPS	2"	
Pressure class	class	class 600	
Maximum operating pressure	p,max	105.0	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	0.923	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	0.887	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	0.573	-
Valve style modifier (Cv/Cv100 = 1)	Fd	0.44	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
s/s100	100.0			%
u2	13.705			m/s
LpAe	91.8			dB(A)
Ri	1.0			-

Noise calculation

Calculation standard (noise, liquid)	IEC 60534-8-4 (2015)
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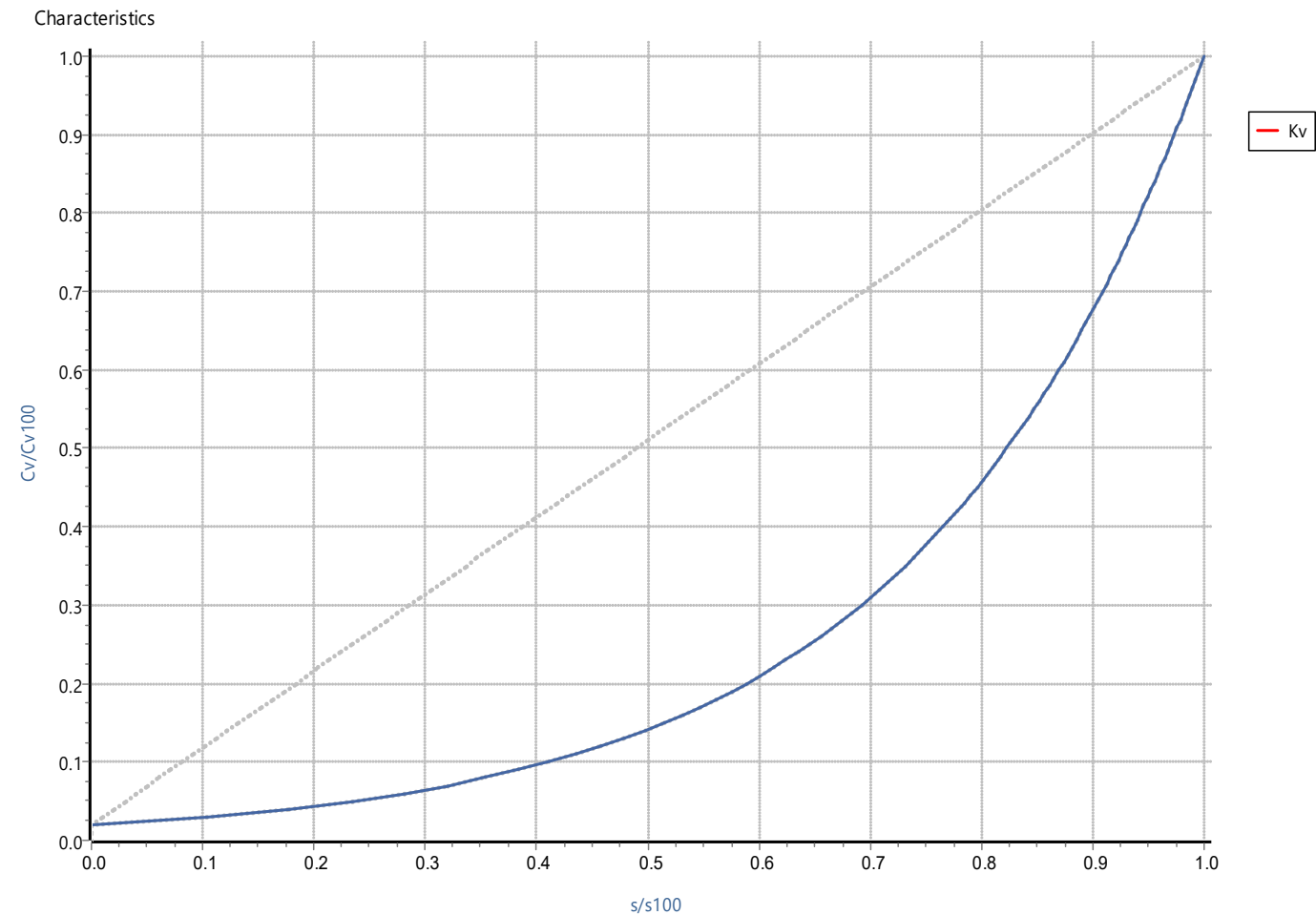




Calculation Sheet

Noise prediction data

Low-noise design		n.a.	
Sound velocity	cF1	1,560.2	m/s
Pressure level	LA1	173.0	dB
Pressure level	LA2	5.57	dB
Sound pressure level	Lpi	178.0	dB
Transmission loss	TL	-81.2	dB
Sound power level (A-weighted)	LwAe	103.0	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	91.8	dB(A)





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Comments:

Reliability index - Ri

Sound pressure level: 20 %; High velocity: 80 %

Recommended solution: Heavy duty valve, Hardened seat/plug, Low-noise

Legend

 Errors

